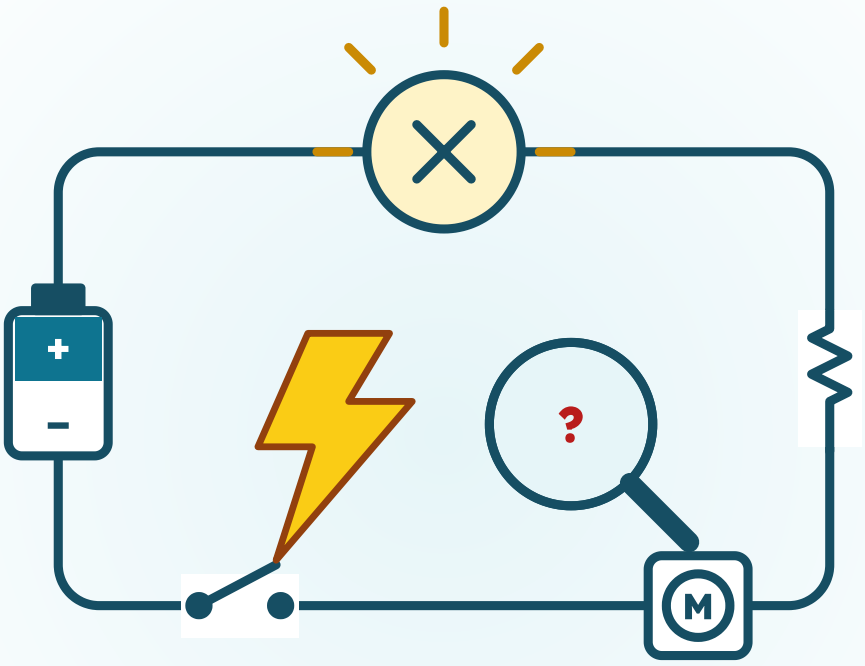
CIRCUIT AGENT

Case File

Trace it. Draw it. Measure it. Fix it.



AGENT ID

Name: _____

Partner: _____

Event: Zap Lab (Diagram & Decode Fridays)

If found, please return to the DDMES Science Olympiad binder shelf.

How a circuit agent works

Welcome, Agent. Electricity is energy that moves through a complete loop called a **circuit**. Your job: crack circuit cases at home, learn the secret symbols, and get ready to beat every station at the tournament with **no notes**.

1
READ
Study the Know Your Event and reference pages (p. 3-6).

2
CRACK
After each meeting, open that meeting's case (p. 7-10).

3
CHECK
Turn to the upside-down answer key on the last page.

4
LOG
Write every practice in your Evidence Log (p. 16).

5
EARN
Show a coach, get initials, color the badge.

The Agent's Code

- [ ] Only use kit batteries (AA, AAA, or 9 V) and the kit parts.
- [ ] Never put anything but a plug in a wall outlet.
- [ ] Never clip a wire straight from + to - (it gets hot).
- [ ] If a battery or wire feels warm, let go and tell an adult.
- [ ] Keep electric things away from water. Dry hands first.
- [ ] Stay far away from power lines, even fallen ones.

My field kit

- [ ] This case file and a pencil
- [ ] A colored pencil for current arrows
- [ ] Your symbol flashcards
- [ ] A flashlight (look only, do not take it apart)
- [ ] A family member or your partner

Journal map

Know your event	p. 3
Reference pages	p. 4-6
Cases 1-4	p. 7-10
Bonus cases	p. 11-13
Drill and sketchbook	p. 14-15
Evidence log	p. 16
Missions, badges, key	p. 17-19

Where to look

Flashlights and toys
a battery (source), a switch, and a load

Battery compartments
+ and - diagrams showing which way to put batteries

Labels and chargers
volts (V) and watts (W) printed on them

The kitchen
loads that make heat, light, sound, and motion

Cords
metal inside (conductor), plastic outside (insulator)

Outside
power lines, streetlights, solar panels

Did you know? The energy travels through a wire almost as fast as light, so a lamp turns on the moment you close the switch!

Know your event: Zap Lab

Big idea: Electricity is energy that moves through a complete loop called a circuit. In Zap Lab you build, draw, measure, and fix small battery circuits, then show what you know at stations.
Tournament format: Pairs rotate through timed stations for up to 50 minutes (sorting materials, reading circuit drawings, looking at real low-voltage circuits and meters, answering questions) and bring only pencils, no notes. Team size: 2 agents.

The rules

- ☐ Bring only a pencil. No notes, no flashcards.
- ☐ Stay at each station until time is called.
- ☐ You may not skip ahead or go back.
- ☐ Whisper only with your partner.
- ☐ Some stations have real low-voltage circuits and meters on the table.

Station smarts

- About 3 minutes per station. Stuck? Skip it and come back if time is left.
- Split the work: one reads, one writes.
- For every circuit, trace the loop with your finger from + back to -.
- Use the Detective checklist (p. 6), one thing at a time.
- Write units! 1.5 **V**, not just 1.5.

Stations on Practice Test 1 (the real format)

1	Energy all around us	**6**	One wire is in the wrong place	**11**	Current and voltage
2	Conductor or insulator?	**7**	Circuit symbols	**12**	Units, resistors, and LEDs
3	Test it	**8**	Draw the schematic	**13**	Read the meter
4	Parts of a circuit	**9**	Read the schematic	**14**	Circuit detective
5	Open or closed?	**10**	Series and parallel	**15**	Safety

Units to know cold

Word	Unit	What it means
Voltage	**V (volts)**	The push that gives current energy to move. One AA = 1.5 V.
Current	**A (amps)**	How much electricity flows past a point.
Resistance	**Ω (ohms)**	How much something slows current. More ohms = less current.
Power	**W (watts)**	How fast a device uses energy.

Can-do checklist: check each one you can do with no notes

- ☐ Name sources of electricity (batteries, outlets, power plants, generators, solar panels) and say that electricity is energy that powers devices.
- ☐ Sort materials into conductors and insulators and explain the difference.
- ☐ Name the parts of a circuit: power source, wires, switch, load.
- ☐ Explain open vs closed circuits and what a switch does.
- ☐ Draw and read symbols for cell, battery, wire, lamp, switch, resistor, LED, motor, buzzer, meter.
- ☐ Predict from a picture or schematic whether a bulb will light, and say why.
- ☐ Build and compare series and parallel circuits.
- ☐ Describe how current flows around a complete loop, from the battery, through the load, and back.
- ☐ Explain voltage as the push and match units: volts (V), amps (A), ohms (Ω), watts (W).
- ☐ Explain that resistance slows current and that an LED (a diode) works only one way.
- ☐ Read a multimeter or meter scale and say what the number means.
- ☐ Troubleshoot a broken circuit: gap, open switch, dead battery, burned-out bulb, backwards LED.
- ☐ Name everyday uses of electricity and the energy each makes (light, heat, sound, motion).
- ☐ Explain and follow electrical safety rules.
- ☐ Answer station questions on diagrams, models, and real situations with no notes.

Secret symbols: the schematic code

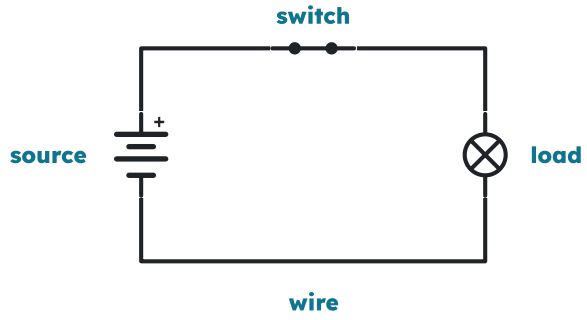
Scientists draw circuits with simple symbols instead of pictures. The drawing is called a **schematic** (skee-MAT-ik). At the tournament you read schematics with no notes, so copy each symbol twice.

Symbol	Name and how to draw it	Copy it	Copy it again
	**Cell** One long line (+) and one short line (-) side by side. One AA battery is one cell.		
	**Battery** Two or more cells joined together, drawn as long-short-long-short lines.		
	**Wire** A straight line. Wires meet with a dot where they join.		
	**Lamp (bulb)** A circle with an X inside.		
	**Switch (open)** A line with a gap and a short line tilted up, like a gate.		
	**Resistor** A zigzag line (or a small rectangle).		
	**LED (diode)** A triangle pointing at a line, with two small arrows coming out (light).		
	**Motor** A circle with an M inside.		
	**Buzzer** A half circle (dome) on a line.		
	**Meter** A circle with V (voltmeter) or A (ammeter) inside.		

Agent tip: the **long** line of a cell is **+**. On an LED, the triangle points the way current flows. Long LED leg = +.

Circuit basics

Parts of a circuit



Power source (gives energy), wires (the path), switch (opens or closes the path), load (uses the energy, like a bulb or motor).

Open or closed?



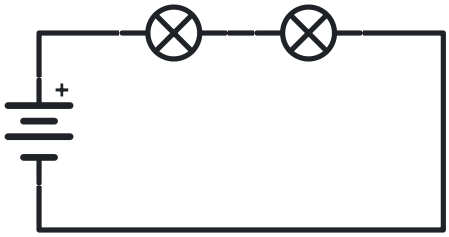
Closed: complete loop. Lamp ON.



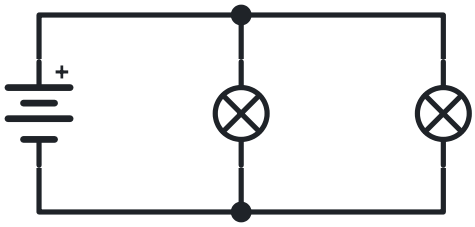
Open: a gap. Lamp OFF.

A **switch** opens and closes the path. Current flows out of **+**, through the load, and back to **-**.

Series or parallel?



Series: One path. Current goes through every part in order. One break stops everything. Bulbs share the push, so they are dimmer.



Parallel: Two or more paths. Each branch works on its own. Bulbs stay bright. Homes and schools are wired this way.

Conductors and insulators

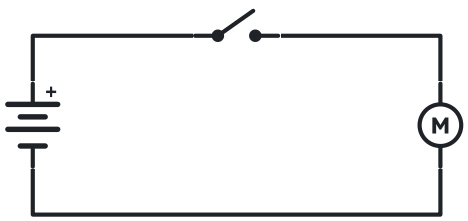
Conductor: A material that lets electricity flow easily. Most metals.

Examples: copper wire, foil, coins, keys, paper clips, steel nails, a silver ring.

Insulator: A material that blocks electricity. Plastic, rubber, wood, glass, paper, cloth.

Examples: rubber band, craft stick, plastic spoon, glass marble, eraser, cotton, string.

Sources of electricity: Batteries, wall outlets (fed by power plants), generators, solar panels.



Quick check

1. Label the source, the switch, and the load on the drawing.

2. Is this circuit open or closed? Will the motor spin?

3. What kind of energy does a motor make?

Safety rules and the Detective checklist

Outlet electricity is much stronger than a battery and can hurt you. Zap Lab agents only use small kit batteries.

Rule	What to do
Outlets and water	Only plugs go in outlets. Keep electric things away from water; dry your hands first. Water and people are conductors.
Cords	Pull the plug, not the cord. Tell an adult about damaged cords.
Power lines	Stay far away. Never touch a fallen line. Fly kites in open fields.
Short circuit	A wire straight from + to - with no load makes a battery hot.

Detective checklist

A circuit will not work? Check **one thing at a time**:

1. Is the battery good? Use the meter.
2. Is every clip on metal?
3. Is the loop complete?
4. Is the switch closed?
5. Is the bulb burned out, or the LED backwards?
6. Is an insulator in the path?

Where outlet electricity comes from

Power plants: Big buildings where generators spin to make electricity.

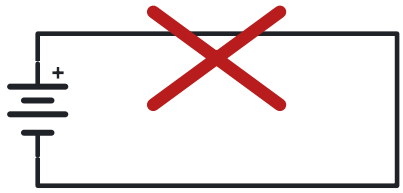
Generators: Machines that turn motion into electricity. Steam, water, or wind can spin them.

Solar panels and wind turbines: Make electricity from sunlight and wind.

Power lines: Carry electricity from the plant to homes, schools, and outlets.

Batteries: Store energy in chemicals and give a small, safe push.

Never build this: a short circuit



A wire straight from + to - with **no load** makes the battery and wire hot. Always put a load (lamp, motor, buzzer, or resistor) in the loop.

My safety pledge

- ☐ I only use kit batteries for Zap Lab.
- ☐ I never touch a fallen power line and I stay far away.
- ☐ I keep electric things away from water.
- ☐ If something gets warm, I let go and tell an adult.

Agent signature: _____

Why cords have two layers: the metal inside is a conductor that carries the current. The plastic outside is an insulator that keeps the current in the wire and keeps people safe.

Case 1: The Sneaky Conductors TOP SECRET

Open this case: after Lesson 1 (Fri Oct 16) · **Badge:** Conductor Detective

Your mission: find these 12 things at home (swap in something close if you can't find one). Look and touch only; **no batteries needed**. Write what each is made of, circle **C** (conductor) or **I** (insulator), and give your clue. Check your answers on p. 19.

Object	Made of	C or I?	My clue (how I know)
1. metal spoon		C / I	
2. plastic cup		C / I	
3. house key		C / I	
4. rubber band		C / I	
5. aluminum foil		C / I	
6. wooden spoon		C / I	
7. coin		C / I	
8. glass jar		C / I	
9. paper		C / I	
10. eraser		C / I	
11. paper clip		C / I	
12. cotton sock		C / I	

Diagram hunt: a flashlight

Look at a flashlight or a battery toy with an adult (do not take it apart). Find each part of the circuit.

Power source	
Switch	
Load	
Path (wires)	

Think like an agent

What do all the conductors have in common?

Why does a wire need both metal and plastic?

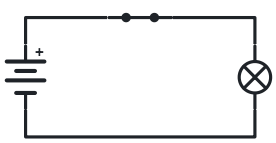
Case 2: The Dark Bulbs

TOP SECRET

Open this case: after Lesson 2 (Fri Nov 20) · **Badge:** Loop Master

Your mission: an agent left 8 circuit drawings on your desk. For each one, trace the loop with a colored pencil from **+** around to **−**. Does the load work? If not, write where the problem is. **G:** write which lamp lights.

A

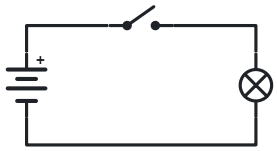


Works? _____

Why? _____

_____

B



Works? _____

Why? _____

_____

C

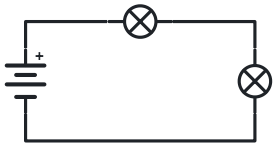


Works? _____

Why? _____

_____

D

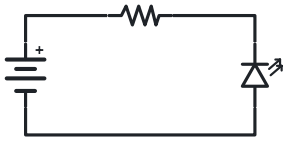


Works? _____

Why? _____

_____

E

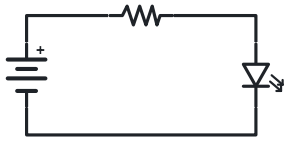


Works? _____

Why? _____

_____

F

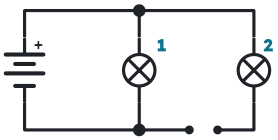


Works? _____

Why? _____

_____

G

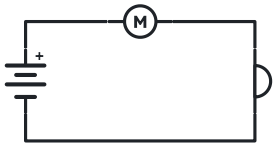


Works? _____

Why? _____

_____

H



Works? _____

Why? _____

_____

Series or parallel?

1. Drawing D has two lamps on one path. Series or parallel?	
2. Drawing G gives each lamp its own branch. Series or parallel?	
3. In D, unscrew one lamp. What does the other lamp do?	
4. Homes and schools are wired in _____. Why?	

Draw it: a battery, a closed switch, and 2 lamps in PARALLEL. Add current arrows.

Case 3: The Mystery Meter TOP SECRET

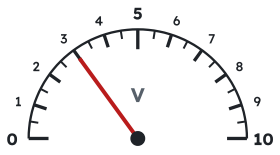
Open this case: after Lesson 3 (Fri Jan 15) · **Badge:** Meter Reader

> **Your mission:** a meter measured 4 batteries. Each small mark is **0.5 V**. Read each needle, then answer the questions. Remember: one fresh AA is about 1.5 V.



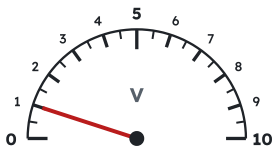
Battery 1

Reading: _____ V



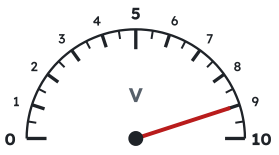
Battery 2

Reading: _____ V



Battery 3

Reading: _____ V



Battery 4

Reading: _____ V

1. Which battery is a used AA that is running low? How do you know?	
2. Which battery could be 2 AA cells in series (1.5 V + 1.5 V)?	
3. Which battery is probably a 9 V battery?	
4. Four 1.5 V batteries in series make how many volts?	

Match the unit

Draw a line from each unit to what it measures.

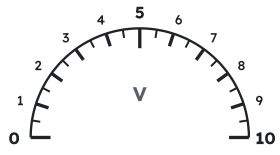
V (volts)	how fast energy is used
A (amps)	the push
Ω (ohms)	how much current flows
W (watts)	how much something slows current

Label hunt at home

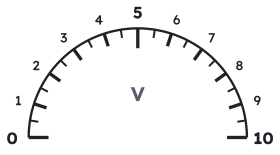
Find 4 labels (batteries, a phone charger, a light bulb box, a lamp or appliance). Copy the number **and** unit.

Where I found it	Number + unit

Your turn: draw the needle



Draw it at 4.5 V



Draw it at 7 V

LED puzzle: an LED and resistor are hooked to a battery, but the LED stays dark. Name two things to check.

_____

_____

_____

Case 4: The Secret Blueprints TOP SECRET

Open this case: after Lesson 4, the mock (Fri Feb 12) · **Badge:** Describe & Draw

Your mission: practice for tournament stations like *Draw the schematic* and *Read the schematic*. Work with your partner or a family member. **Agent A** folds the page on the dashed line and looks only at Secret 1. Agent A describes it in words (no pointing, no drawing in the air!). **Agent B** draws it in Box 1. Then switch jobs for Secret 2.

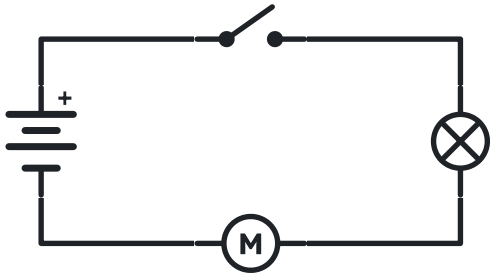
Box 1: Agent B draws Secret 1 here

Box 2: Agent A draws Secret 2 here

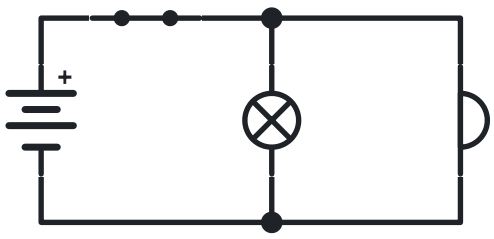
Check it	Box 1	Box 2
Every part is there (1 point each)		
Parts are in the right order and place (2 points)		
Switch drawn open or closed correctly (1 point)		
Words that helped most (like "series", "branch", "on the left side")		

✂ - - - - - - - - fold here: the drawer must not see below this line - - - - - - - - ✂

SECRET 1 (Agent A only)



SECRET 2 (Agent B only)



Bonus Case: The Six Broken Circuits

TOP SECRET

Open anytime after Lesson 3 · **Badge:** Circuit Fixer

Your mission: six circuits stopped working. Read each clue and pick the culprit from the lineup. Then write how to fix it.

The lineup of suspects:

dead battery burned-out bulb insulator in the path LED backwards switch open gap (open circuit)

	Clue	Culprit	How to fix it
1	**Case of the tired battery** The meter reads 0.4 V on the AA battery. Everything else looks fine.		
2	**Case of the broken filament** The loop is complete, the battery reads 1.5 V, but the tiny wire inside the bulb is snapped.		
3	**Case of the plastic clip** One clip lead is clamped onto the plastic coating of a wire, not the metal.		
4	**Case of the dark LED** The LED's long leg goes toward the - end of the battery.		
5	**Case of the lazy gate** The drawing of the switch shows the gate tilted up.		
6	**Case of the loose end** One clip lead is attached at the bulb, but its other end is lying on the table.		

From memory: the Detective checklist

Cover page 6. Write the 6 things to check, in order.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

Why one thing at a time? If you change 3 things and it works, you won't know which one was the real culprit.

Bonus Case: The Energy Hunt TOP SECRET

Open anytime · **Badge:** Energy Spotter

> **Your mission:** electricity is energy that powers devices. A **load** changes it into light, heat, sound, or motion. Search your home, school, and town. Find 2 devices for each kind of energy.

Energy made	Device	Home, school, or town?	Battery or outlet?
Light			
Light			
Heat			
Heat			
Sound			
Sound			
Motion			
Motion			

Two-energy trick

Many loads make more than one kind. A hair dryer makes heat, motion (the fan), and sound. Find one device that makes 2 or more kinds.

_____

_____

_____

Source detective

Name 3 sources of electricity. Circle the one that powers most of your house.

_____

_____

_____

Draw a schematic of one device you found (source, switch, load)

Bonus Case: The Home Safety Sweep

TOP SECRET

Open anytime (goes with Lab I3) · **Badge:** Safety Captain

> **Your mission:** walk through your home with an adult. **Look only, do not touch or unplug anything yourself.** Check each thing that is safe. Write down anything to fix and show your adult.

Kitchen	☐ appliances far from the sink ☐ toaster unplugged when not used ☐ no cords near the stove
Bathroom	☐ hair dryer unplugged and away from water ☐ dry hands before touching switches
Bedroom and living room	☐ no cords under rugs ☐ outlet strip not overloaded ☐ no damaged cords ☐ unused outlets covered (if little kids)
Outside	☐ know where power lines are ☐ a kite-flying spot far from lines ☐ family knows: stay away from fallen lines

Things we should fix

Where	What we saw	How we fixed it

Quick quiz: safe or unsafe?

1. Flying a kite near power lines	safe / unsafe
2. Using a hair dryer next to a full bathtub	safe / unsafe
3. Putting a fork in a toaster	safe / unsafe
4. Touching a 9 V battery to your tongue	safe / unsafe

If a power line is down

Write what your family will do. Who will you tell? Where will you stay?

> **Water and people are conductors.** That is why electric things and water never mix.

Symbol Memory Drill

Cover page 4. Draw each symbol from memory. Then check it and mark any you missed. Do all 10 right in one try to earn the **Symbol Master** badge.

Symbol	Try 1 Date: _____	Try 2 Date: _____	Try 3 Date: _____
Cell			
Battery			
Wire			
Lamp (bulb)			
Switch (open)			
Resistor			
LED (diode)			
Motor			
Buzzer			
Meter			
Right	____ / 10	____ / 10	____ / 10

Schematic Sketchbook

Turn real things into schematics. Use the symbols from page 4. Ideas: a flashlight (2 cells, switch, lamp), a battery fan, a doorbell, a string of holiday lights, a toy car.

Device: _____

Device: _____

Device: _____

Device: _____

Series or parallel? How can you tell?

What happens if one part breaks?

Evidence Log: my practice

Every time you practice (flashcards, a case, a lab, the practice test), write it here. 6 sessions earns the Study Streak badge.

#	Date	What I practiced	Minutes	Score or how many right	Still tricky
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					
25					

Case Missions

One case for each Zap Lab meeting. Finish it, show a coach or your Team Captain for initials, then color the badge.

When	Mission	Badge	Done	Initials
After Fri Oct 16 Lesson 1	Case 1: sort 12 home objects and find the parts of a flashlight (p. 7).	Conductor Detective	☐	
After Fri Nov 20 Lesson 2	Case 2: crack all 8 Will it light? drawings (p. 8).	Loop Master	☐	
After Fri Jan 15 Lesson 3	Case 3: read 4 meters and find 4 labels (p. 9).	Meter Reader	☐	
After Fri Feb 12 Lesson 4	Case 4: Describe & Draw both secret schematics (p. 10).	Describe & Draw	☐	
Anytime Bonus	Draw all 10 symbols from memory (p. 14). Solve the 6 broken circuits (p. 11).	Symbol Master / Circuit Fixer	☐	
Anytime Bonus	Energy hunt (p. 12) and home safety sweep (p. 13).	Energy Spotter / Safety Captain	☐	
All season Log	Log 6 practice sessions in your Evidence Log (p. 16).	Study Streak	☐	

Tournament day: Sat Mar 13

- ☐ 2 sharp pencils (no notes allowed!)
- ☐ A good night's sleep
- ☐ Review your 2 weakest topics from the mock the night before

- Trace every loop with your finger from + back to -.
- Write the unit with every number (V, A, Ω, W).
- Stuck? Skip it, then come back if time is left.

My Badges

Color in each badge when you earn it. Earn all 9 to become a **Master Circuit Agent**! Your Team Badge Passport has the same badges.



Conductor Detective

Sort 12 home objects (Case 1)

Initials: _____



Loop Master

Crack all 8 Will it light? drawings (Case 2)

Initials: _____



Meter Reader

Read 4 meters and 4 labels (Case 3)

Initials: _____



Describe & Draw

Draw a secret schematic from words (Case 4)

Initials: _____



Symbol Master

Draw all 10 symbols from memory

Initials: _____



Circuit Fixer

Solve the 6 broken-circuit clues

Initials: _____



Energy Spotter

Find light, heat, sound, and motion

Initials: _____



Safety Captain

Do a home safety sweep

Initials: _____



Study Streak

Log 6 practice sessions

Initials: _____

Turn the book upside down to read the answers.

Answer key

Agents: finish the case first, then check. Fix mistakes in a different color.

Case 1: The Sneaky Conductors

Conductors (C): metal spoon, house key, aluminum foil, coin, paper clip. Insulators (I): plastic cup, rubber band, wooden spoon, glass jar, paper, eraser, cotton sock. The clue: conductors are metal. Flashlight: source = batteries, switch = the button, load = the bulb, path = metal strips or wires inside. A wire needs metal to carry the current and plastic to keep it in and keep people safe.

Case 2: The Dark Bulbs

A. works. Complete loop, switch closed.

B. no. The switch is open.

C. no. There is a gap in the bottom wire (open circuit).

D. yes, both lamps light (dimmer). One path, series.

E. no. The LED is backwards (it works only one way).

F. yes. The LED points the right way and the resistor protects it.

G. lamp 1 lights. Lamp 2's branch has a gap, so only lamp 2 is dark.

H. yes. Motor spins and buzzer buzzes (series).

1 series. 2 parallel. 3 it goes out too (one break stops everything in series). 4 parallel, so each light works on its own and stays bright.

Case 3: The Mystery Meter

Battery 1 = 1.5 V (fresh AA). Battery 2 = 3 V. Battery 3 = 1 V. Battery 4 = 9 V. 1) Battery 3: 1 V is less than 1.5 V. 2) Battery 2. 3) Battery 4. 4) 6 V. Match: V = the push, A = how much current flows, Ω = how much something slows current, W = how fast energy is used. LED puzzle: is the LED backwards? Is the battery dead? Is every clip on metal? Is the loop complete?

Case 4: The Secret Blueprints

Secret 1: battery, open switch, lamp, and motor, all in one loop (series). Nothing runs because the switch is open. Secret 2: battery and closed switch, then 2 branches in parallel: a lamp and a buzzer. Both work.

Bonus: The Six Broken Circuits

1 dead battery (replace it). 2 burned-out bulb (new bulb). 3 insulator in the path (clip onto the metal). 4 LED backwards (flip it: long leg to +). 5 switch open (close it). 6 gap (clip the loose end back to the battery).

Bonus: Safety quiz

Kite near power lines: unsafe. Hair dryer by a full tub: unsafe. Fork in a toaster: unsafe. 9 V battery on your tongue: unsafe.

Circuit basics quick check (p. 5)

1. Source = the battery, switch = the gate on the top wire, load = the motor. 2. Open, so the motor will not spin. 3. Motion.